UNIT 0 · THE PHYSICIST'S TOOLKIT · SKILLS CHECK RESOURCE

How to Build a Cheat Sheet That Actually Helps

Physics 2026–27 · your first card: **one side of one 4×6 index card** · due at the Skills Check, Monday, September 14

> **The policy.** One side of one 4×6 index card. **Handwritten by you** — nothing typed, printed, photocopied, or borrowed. Equations, definitions, triggers, and reminders only — **no worked problems**. The card comes with you to every Skills Check and test this year, and the rules get stricter as you earn them: some future checks will shrink the card. Building it well is the studying — here's how.

Why the card exists

A cheat sheet is not a place to store answers — it's a machine that forces your brain to do three things it hates: **retrieve** (recall what the unit covered), **judge** (decide what you actually struggle with), and **compress** (shrink each idea until it fits). By the time the card is done, most of what's on it you'll barely need — and that's the point. Students who copy a friend's card or print one out skip all three steps and bring a luggage cart of text they'll never find under time pressure.

Handwritten vs. typed

	HANDWRITING	TYPING
SELECTION	**The feature, not the bug.** A 4×6 card runs out of space fast, so you're forced to triage — and triage is where the learning happens.	There's no natural limit, so cards bloat: 7-point font, twelve columns, and on test day you can't find anything in it.
MEMORY	Writing by hand forces a decision per line, and the *position* of each fact on the card becomes its own memory cue ("friction is bottom-left, next to μ").	Fast fingers let you paste without deciding. Feels productive; encodes far less.
SPEED	Slow to produce, slower to revise. Redraft on scrap paper first so you're not inking a card five times.	Fast to produce and infinitely easy to edit, reorder, and reuse for the next unit.
THIS CLASS	**Required for the card you bring.**	Legal and smart for the *brainstorm* stage only (see the ritual below).

The hybrid move: type your brain-dump, delete ruthlessly until it fits a 4×6 box on screen, *then* hand-write the survivors onto the card. The typed list is scaffolding; the hand-copying is one more pass of retrieval before the check.

What makes the cut

1. **What *you* keep forgetting.** Go back through PS1–PS3 and the study guide: every miss, every correction, every "wait, which one divides?" gets priority. What you already know cold does not belong on the card — it's spending space you don't have.
2. **Triggers and cues, not sentences.** "exponents first!" beats "when comparing two numbers written in scientific notation, begin by comparing their exponents." The card cues your memory; it doesn't replace it.
3. **Rules with the reason compressed out.** "answer w/o unit = no credit" — five words carry a whole class period.
4. **Equations and conversion factors, units attached.** A formula without its units is half a formula.
5. **Decoders and warnings.** The Greek alphabet, the DEG-mode check, the ≈-versus-= rule — the small stuff that costs big points.

Leave off: worked problems (banned, and useless — you'd need a new one anyway), definitions you can re-derive in one line, full sentences, anything you can't explain out loud. If you can't explain a line on your own card, that line is a liability — know every line you bring.

The build ritual

1. **Brain-dump (10 min).** On scrap paper, list everything Unit 0 covered, fast, no filter. Slower than the memory, the page — write until it stops coming.
2. **Triage.** Cross off what you know cold. Star what you got wrong or looked up on PS1–PS3 and the study guide. The stars write first.
3. **Compress.** Shrink each survivor to a cue of five words or fewer. Say it out loud as a full sentence to check it still triggers the whole idea.
4. **Zone the card in pencil.** Title across the top, three columns, one strip along the bottom. Pencil first — one wasted card is fine; two is triage you skipped.
5. **Ink it.** Biggest ideas biggest. Units in every formula. Nothing crammed at the bottom edge.
6. **Card test (below).** Then adjust — space you never touched is space someone else deserved.

Unit 0: things to know (checklist)

Everything below is fair game Monday. Check items as you place them — or as you consciously decide they're already automatic.

- ☐ SI base units and symbols: mass `kg`, time `s`, length `m` — and which common unit is *derived* (`N = kg·m/s²`)
- ☐ Scientific notation: coefficient 1–10 × 10^n, converting both directions
- ☐ Sig figs: leading zeros never, trapped zeros always, trailing only with a decimal, `1200`-style ambiguity
- ☐ Prefix powers: `G` 10^9 · `M` 10^6 · `k` 10^3 · `c` 10^{-2} · `m` 10^{-3} · `μ` 10^{-6} · `n` 10^{-9}, and which way the number moves in each conversion
- ☐ Comparing powers of ten without writing them out
- ☐ KDEA, labeled, on every quantitative problem

- ☐ Units travel with the math; an answer without a unit does not count
- ☐ ≈ for rounded, = only when exact
- ☐ Opposite/adjacent flip when you switch angles; hypotenuse never moves
- ☐ Pythagorean theorem, both directions: unknown hypotenuse *and* unknown leg
- ☐ A vector is magnitude AND direction — both halves or half credit
- ☐ The Greek decoder: Δ Σ θ ω λ μ ρ τ
- ☐ Fermi estimation: 3–4 guessables, written down, multiplied to an order of magnitude
- ☐ Symbols rearranged before numbers go in
- ☐ SOH-CAH-TOA — and how to *pick* the ratio (side you know + side you want)
- ☐ Inverse trig for angles: $\theta = \tan^{-1}(\text{opp/adj})$
- ☐ Components: $x = r \cos\theta$, $y = r \sin\theta$
- ☐ The 3-4-5 family and its ≈37°/53° angles
- ☐ Calculator in DEG mode — the $\sin 30° = 0.5$ check

Compression drill

Practice the shrink. Each row is a real sentence a student wanted to bring, and the cue that does the same job in a fraction of the space.

	CARD HOG (DON'T BRING)	CUE (BRING THIS)
SCI. NOTATION	"When comparing two numbers in scientific notation, you should always look at the exponent before doing anything else with the numbers."	compare: EXPONENTS first!
HYPOTENUSE	"The hypotenuse is always the longest side of a right triangle and it is always located across from the right angle."	hyp: longest · opposite 90° · never moves
CONVERSIONS	"To convert kilometers to meters you multiply by one thousand because the prefix kilo means one thousand."	k = ×10³ (k grows!)

The example card

Below is a strong Unit 0 card — one side, hand-written-layout, everything from the checklist placed *once*. Print this page at 100% and the dashed box is a true 4×6 in: hold your real index card against it. That rectangle is your entire budget.

ONE SIDE · THE WHOLE CARD

Unit 0 — The Physicist's Toolkit

UNITS & PREFIXES

- SI: mass `kg` · time `s` · length `m`
- force → N = `kg·m/s²` (derived!)
- answer w/o unit = no credit
- k 10^{3} · M 10^{6} · G 10^{9} (grow!)
- c 10^{-2} · m 10^{-3} · μ 10^{-6} · n 10^{-9}
- sci. not.: 1–10 × 10^{n}
- compare: exponents first!

KDEA & NUMBERS

- **K** knowns w/ units · **D** target + unit
- **E** equation · **A** symbols first
- units travel the whole way
- sig figs: leading 0s never · trapped always
- trailing only w/ decimal
- 1200 ambiguous → sci. not.
- ≈ rounded · = exact

TRIANGLES & VECTORS

- sin = opp/hyp · cos = adj/hyp · tan = opp/adj
- pick ratio w/ KNOWN + WANT
- opp/adj flip w/ angle · hyp never moves
- no angle: $a^2+b^2=c^2$ · θ = $\tan^{-1}$(opp/adj)
- components: x = r cosθ · y = r sinθ
- vector = magnitude AND direction
- 3-4-5 family → ≈37°/53°

GREEK · CALCULATOR · FERMI

Δ change · Σ sum · θ angle · λ wavelength · ω ang. vel. · μ friction (or micro!) · ρ density · τ torque | **DEG mode!** sin 30° must = 0.5 | Fermi: 3–4 guesses → order of mag

Notice the choices: every line is a cue, not a sentence; no worked problems anywhere; each idea appears once; the bottom strip holds the small stuff that costs big points. *Your card should not match this one.* Yours should have gaps where you're already solid and extra ink where you keep slipping — this example just proves everything fits.

The card test

1. Sit down with the finished card and work **Study Guide problems 7 and 11** fresh — no other notes.
2. Every time you reach for the card, tally it. Anything you looked for and *didn't find* goes on before Monday.
3. Anything you never touched is wasting space — erase it and give that real estate to a starred item.
4. Last check: read the card top to bottom and explain every line out loud in one sentence. A line you can't explain is a line you can't use.

Bring the card Monday, September 14 — handwritten, one side, 4×6. It rides with you the rest of the year.